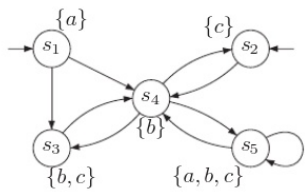
LTL

1. Question 1 on LTL is as follows.

EXERCISE 5.2. Consider the transition system TS over the set of atomic propositions $AP = \{a, b, c\}$:



Decide for each of the LTL formulae φ_i below, whether $TS \models \varphi_i$ holds. Justify your answers! If $TS \not\models \varphi_i$, provide a path $\pi \in Paths(TS)$ such that $\pi \not\models \varphi_i$.

$$
\begin{aligned}
\varphi_1 &= \Diamond \Box\, c \\
\varphi_2 &= \Box \Diamond\, c \\
\varphi_3 &= \bigcirc \neg c \rightarrow \bigcirc \bigcirc\, c \\
\varphi_4 &= \Box\, a \\
\varphi_5 &= a \,\mathsf{U}\, \Box\, (b \vee c) \\
\varphi_6 &= (\bigcirc \bigcirc\, b) \,\mathsf{U}\, (b \vee c)
\end{aligned}
$$

2. Question 2 on LTL is as follows.

EXERCISE 5.4. Suppose we have two users, *Peter* and *Betsy*, and a single printer device *Printer*. Both users perform several tasks, and every now and then they want to print their results on the *Printer*. Since there is only a single printer, only one user can print a job at a time. Suppose we have the following atomic propositions for *Peter* at our disposal:

- *Peter.request* ::= indicates that *Peter* requests usage of the printer;
- *Peter.use* ::= indicates that *Peter* uses the printer;
- *Peter.release* ::= indicates that *Peter* releases the printer.

For *Betsy*, similar predicates are defined. Specify in LTL the following properties:

(a) Mutual exclusion, i.e., only one user at a time can use the printer.

(b) Finite time of usage, i.e., a user can print only for a finite amount of time.

(c) Absence of individual starvation, i.e., if a user wants to print something, he/she eventually is able to do so.

(d) Absence of blocking, i.e., a user can always request to use the printer

(e) Alternating access, i.e., users must strictly alternate in printing.

Answer to Exercise 5.2

$\varphi_1 = \Diamond\Box\, c$	no	$s_2 s_4 s_2 s_4 \ldots$
$\varphi_2 = \Box\Diamond\, c$	yes	
$\varphi_3 = \bigcirc\neg c \rightarrow \bigcirc\bigcirc\, c$	yes	
$\varphi_4 = \Box\, a$	no	$s_2 \ldots$
$\varphi_5 = a \,\mathsf{U}\, \Box\,(b \vee c)$	yes	
$\varphi_6 = (\bigcirc\bigcirc\, b) \,\mathsf{U}\, (b \vee c)$	no	$s_1 s_4 s_2 \ldots$

Answer to Exercise 5.4

(a) $\Box\,\neg\,(Peter.use \;\wedge\; Betsy.use)$

(b) Finite time of usage means that in each state that a user uses the printer, for all possible following computations, at some point in time this user should have released the printer. This should hold for both *Peter* and *Betsy*, of course:

$$\Box\,((Peter.use \;\Rightarrow\; \Diamond\, Peter.release) \;\wedge\; (Betsy.use \;\Rightarrow\; \Diamond\, Betsy.release))$$

(c) $\Box\,((Peter.request \;\Rightarrow\; \Diamond\, Peter.use) \;\wedge\; (Betsy.request \;\Rightarrow\; \Diamond\, Betsy.use))$

(d) If a user requests access to the printer, it should be impossible to be in this *request* state forever:

$$\Box\,((Peter.request \;\Rightarrow\; \neg\Box\, Peter.request) \\ \wedge\; (Betsy.request \;\Rightarrow\; \neg\Box\, Betsy.request))$$

(e) Strict alternating access to the printer can be enforced by expressing that if a user is using the printer he/she will use it until he/she will not use it until the other user has used it.

$$\Box\,((Peter.use \;\Rightarrow\; Peter.use \,\mathsf{U}\, (\neg\, Peter.use \,\mathsf{U}\, Betsy.use)) \\ \wedge\; Betsy.use \;\Rightarrow\; Betsy.use \,\mathsf{U}\, (\neg\, Betsy.use \,\mathsf{U}\, Peter.use)))$$

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